



# Long Term Disease Treatment Monitoring and Care Coordination System

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## 1. Abstract

A healthcare platform to manage and monitor the treatment journey of patients with long-term and serious diseases. It organizes patients, diseases, treatment plans, medications, tests, appointments, symptoms and follow-ups, so providers can track progress and patients can view schedules and report side effects.

**Main Outcome:** Reliable, organized platform improving continuity of care and provider–patient coordination.

## 2. Problem Statement & Objectives

**Problem:** Long-term treatment involves many medicines, tests and visits. Scattered records make progress hard to track and follow-ups easy to miss.

- Obj 1:** Design a structured database for patients, diseases and treatment plans.
- Obj 2:** Build REST APIs with Node.js/Express and integrate the database with the live backend.
- Obj 3:** Track medications, tests, appointments and symptom reports with notifications.

**Scope:** Patients and healthcare providers. **Boundary:** web-based care coordination system.

## 3. Existing vs Proposed System

| Feature            | Existing                                                  | Proposed                       |
|--------------------|-----------------------------------------------------------|--------------------------------|
| Treatment Records  | Paper / scattered files                                   | Centralized digital plans      |
| Medication & Tests | Manual tracking                                           | Scheduled, tracked over time   |
| Appointments       | Phone / manual                                            | Booking, reminders, follow-ups |
| Symptoms           | Reported only at visits                                   | Patient logs anytime           |
| Gap                | Poor continuity of care → organized, coordinated platform |                                |

## 4. Feasibility Study

**Technical** React, Node/Express and MongoDB, all open-source and well documented.

**Economic** Free tools (Git, GitHub, Postman); low development cost.

**Operational** Simple dashboards for patients and providers.

**Schedule** Semester project, Term-III 2025-26.

## 5. Tools & Technologies

**Front-End** React.js, React Router, Context API / Redux

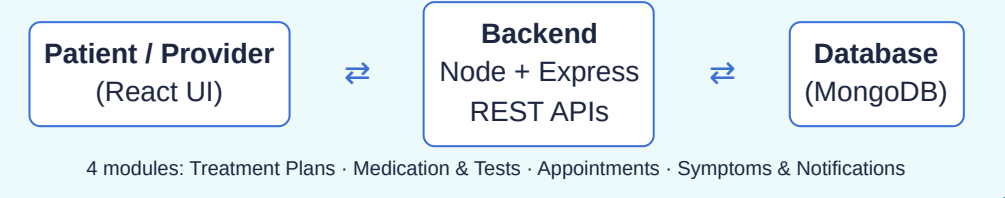
**UI Styling** Tailwind CSS / Material UI

**Back-End** Node.js, Express.js (REST APIs)

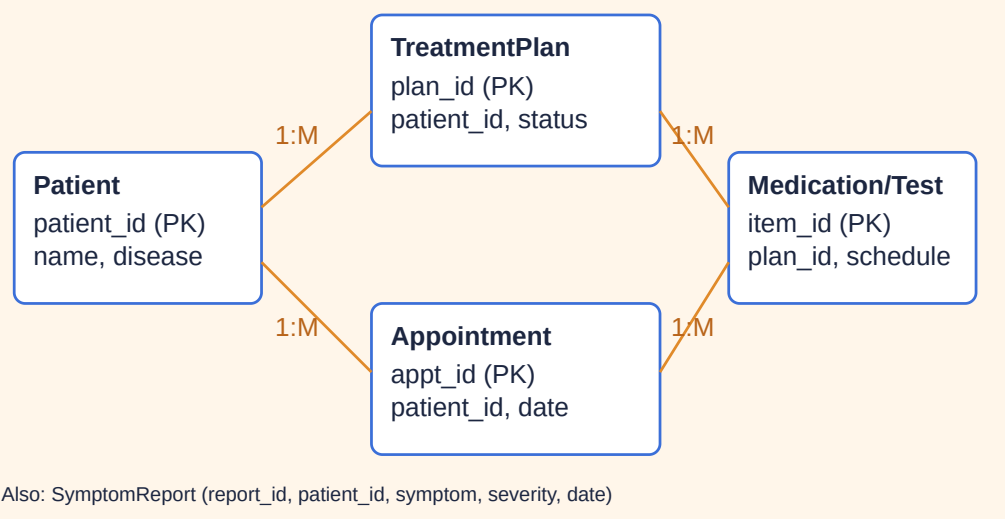
**Database** MongoDB

**Tools** Git, GitHub, Postman, Chart.js / Recharts

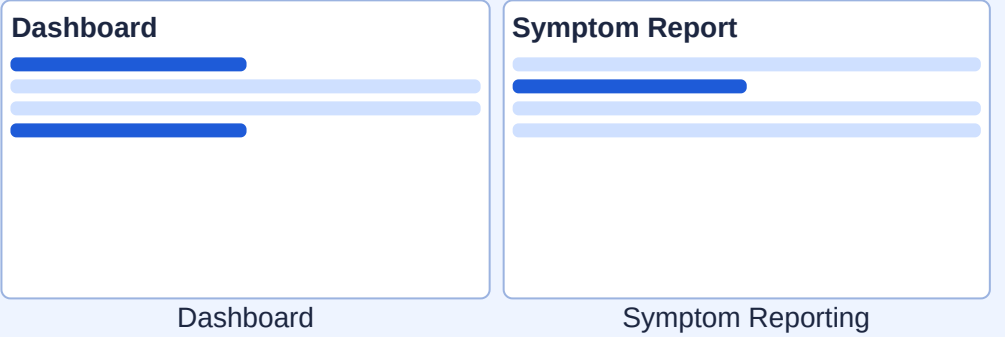
## 6. System Architecture



## 7. ER Diagram & Database Design



## 8. Implementation Screenshots



## 9. Results & Testing

### Results

- Centralized treatment records
- Medication & test tracking
- Appointment reminders
- Symptom reporting dashboard

### Testing

- REST API testing (Postman)
- CRUD operations
- Form validation
- Notification checks

## 10. Conclusion

The system organizes long-term treatment data, improves continuity of care and strengthens coordination between patients and providers.

## 11. Future Scope

- Docker deployment & CI/CD
- Microservices split of modules
- Advanced reminders & analytics

## References

- [1] React.js Docs
- [2] Node.js / Express.js Docs
- [3] MongoDB Docs
- [4] GitHub Docs